IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Date of publication October 00, 2026, date of current version October 00, 2026.

Digital Object Identifier 10.1109/ACCESS.2026.xxxxxxx

DANG: Adapting Frozen DINOv3 Features with a Token-Pyramid Decoder for Generalizable Polyp Segmentation

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ABSTRACT Polyp segmentation networks trained on two colonoscopy centres lose accuracy on frames from other centres and devices. We present DANG, which adapts a frozen DINOv3 ViT-S/16 encoder with convolutional bottleneck adapters, decodes a four-scale token pyramid with a scale-aware decoder, and adds a dual-prototype margin head; only 1.9M of its 23.5M parameters are trained. We train on the standard 1,450-image Kvasir-SEG and CVC-ClinicDB split, develop on the five standard test sets and 1,399 deduplicated PolypGen frames, and hold out 1,376 PolypDB frames as a final test evaluated once. PraNet, Polyp-PVT, PraNet-V2 and SegDINO are retrained from their official code with their own recipes, and every model is scored at its final epoch by one evaluator over five seeds. On the held-out PolypDB, DANG reaches 89.8% mean Dice against 86.7% for the strongest retrained model, Polyp-PVT (paired $p = 0.005$). On the development cohorts it gains 7.3 and 2.7 points on ETIS-LaribPolypDB and PolypGen and shows no significant difference on CVC-ColonDB; it trails Polyp-PVT by 3.2 points on the in-domain CVC-ClinicDB and in boundary F1 on three of the four large external cohorts, and its advantage comes mainly from fewer failed frames. Ablations attribute 5.5–10.8 points on the primary cohorts to the adapters and 0.2–1.0 points, not significant across seeds, to the prototype head. With FP16 weights and CUDA graphs a frame takes 9.4 ms on an NVIDIA T4.

INDEX TERMS Colonoscopy, deep learning, domain generalization, foundation models, medical image segmentation, parameter-efficient fine-tuning, polyp segmentation, prototype learning.

I. INTRODUCTION

COLORECTAL cancer is the third most common cancer and the second leading cause of cancer death worldwide [1], and removing adenomatous polyps during colonoscopy lowers colorectal-cancer mortality [2]. Pixel-level polyp segmentation supports this by delineating lesion margins. The standard benchmark trains on 1,450 frames from two centres, Kvasir-SEG [3] and CVC-ClinicDB, and tests on five public sets [4]. On the two in-domain test sets current CNN and transformer models reach about 90% mean Dice or more; on CVC-ColonDB [5] and ETIS-LaribPolypDB [6], which come from other centres and cameras, these models lose 10–26 points (Table 2). Generalization to unseen centres, not in-domain accuracy, is the open problem.

Vision foundation models such as DINOv2 [7] and DINOv3 [8] are trained on very large image collections and transfer across domains, which makes them natural encoders for this problem. Keeping such an encoder frozen also keeps training cheap and the trainable part small. SegDINO [9] pairs a frozen DINOv3 with a light decoder; under our protocol, however, it reaches only 43.3% mean Dice on ETIS (Table 2). A frozen encoder trained on natural images does not see colonoscopy frames, and its 16-pixel patch grid is coarser than lesion boundaries, so how the frozen features are adapted and decoded decides whether a frozen foundation model helps.

We present **DANG** (Figure 1) and make three contributions:

- **An adapted frozen-foundation-model segmenter.** DANG adapts a frozen DINOv3 ViT-S/16 with convolutional bottleneck adapters, decodes a token pyramid with the scale-aware decoder of the extended SegDINO [33], and adds a dual-prototype margin head with test-time

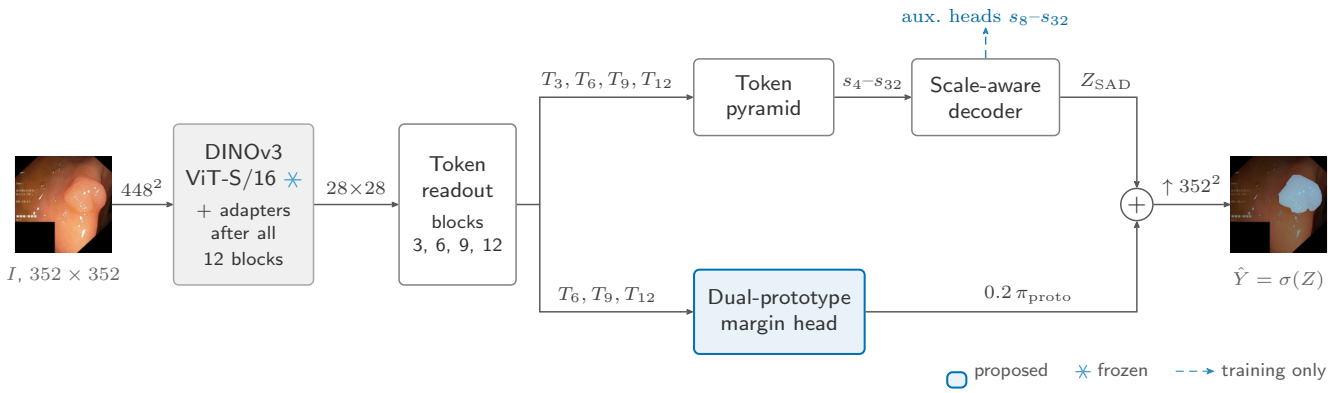


FIGURE 1. DANG adapts a frozen DINOv3 encoder and decodes it through a token pyramid. A 352 × 352 frame is resized to 448 × 448 and encoded by a frozen DINOv3 ViT-S/16 whose 12 blocks are each followed by a trained convolutional bottleneck adapter. Tokens from blocks 3, 6, 9 and 12 are projected to 192 channels on the 28 × 28 grid. The token pyramid assigns one depth to each stride (s_4–s_{32}) and the scale-aware decoder fuses them top-down into the logit Z_{SAD}. The dual-prototype margin head (blue) turns the three deepest maps into the margin map π_{proto}, which is added to Z_{SAD} with weight 0.2. Dashed: heads used only for deep supervision during training. Thumbnails: a Kvasir test frame and its ground-truth mask, for illustration only.

self-support. It trains 1.9M of its 23.5M parameters and needs no convolutional backbone.

- **A controlled benchmark.** The standard training split; the five standard test sets kept as published; two deduplicated multi-centre cohorts, PolypGen [10] and PolypDB [11], the latter held out and evaluated once as a final test; four published models retrained from their official code with their own recipes; final-epoch checkpoints, one evaluator, five seeds, paired tests with multiplicity correction, and boundary and failure-rate metrics next to Dice.
- **Ablations.** Over five seeds we remove the adapters, the prototype head and both, and on the same frozen encoder and token pyramid we compare three decoders and report what each costs in latency on an NVIDIA T4.

II. RELATED WORK

a: Polyp segmentation.

Architectures progressed from U-Net [12] and UNet++ [13] to reverse-attention networks: PraNet [4] mines boundary cues with parallel partial decoders and reverse attention, and CaraNet [14] adds axial context for small objects. Transformer encoders improved multi-centre generalization in Polyp-PVT [15] and TransFuse [16], and EMCAD [17] pairs a PVTv2 encoder with an efficient multi-scale convolutional attention decoder. Recent models refine the decoder further: PraNet-V2 [18] supervises the background explicitly inside reverse attention, ASGNet [19] guides ResNet or PVTv2 features with an adaptive spectral filter, and CSG-Mamba [20] adds convolutional gating to a vision state-space U-Net. Their reported gains are hard to compare: an audit of 27 polyp segmentation papers [21] found at least five incompatible train/test splits of the same datasets, and 26 of the papers reported no significance test. Label-efficient work on polyps has mostly been semi-supervised; FPGM [22], for example, uses a frequency prior on polyp edges to augment unlabelled images and improve generalization to new centres.

b: Foundation models for polyps.

SAM applied without prompts segments polyps poorly [23], and MedSAM [24] fine-tunes SAM on a large medical corpus but requires box prompts. SAM 3 is being adapted in the same way: Medical SAM3 [25] fully fine-tunes it on 33 medical datasets, SAM3-Adapter [26] trains lightweight adapters, and HAT-SAM3 [27] adapts it to colonoscopy with a highlight-aware boundary regulariser. SAM-Mamba [28] and SAM-MaGuP [29] adapt SAM with Mamba-based priors for zero-shot polyp generalization, at about 103M parameters and 423 GFLOPs [30]. A second line distils foundation models into small networks: Polyp-DiFoM [31], LiteBounD [30] and Lite-π [32]. Closest to DANG, SegDINO [9] keeps a DINOv3 encoder [8] frozen and decodes multi-depth tokens with a light head, and its extended version [33] arranges the tokens into a multi-scale pyramid with a scale-aware decoder. DANG builds on this pyramid and decoder, adds trained adapters and a prototype margin head, and keeps the foundation model at inference without distillation.

c: Prototype-based segmentation.

Few-shot semantic segmentation classifies query pixels by their similarity to class prototypes computed from support images [34]. Self-support prototypes [35] are built from confident features of the query image itself, which narrows the gap between support and query appearance. DANG adopts this self-support idea inside a fully supervised segmenter: learned prototypes and ground-truth anchors give a prior, and image-specific self-support centroids adapt it at test time.

d: Adapting frozen vision transformers.

Low-rank adaptation (LoRA [36]) adapts frozen transformers with few parameters; Conv-LoRA [37] adds convolutional experts to the low-rank branch of SAM's encoder, and convolutional bypass adapters such as Convpass [38] add a small convolutional residual path to each block. Dense-prediction heads such as DPT [39] reassemble ViT tokens from several

depths into a multi-scale map. DANG uses a residual convolutional bottleneck after every block of the frozen encoder, in the spirit of these adapters, and a fixed depth-to-stride assignment instead of learned reassembly.

III. METHOD

DANG maps an RGB frame $I \in \mathbb{R}^{3\times352\times352}$ to a polyp probability map $\hat{Y} = \sigma(Z)$ (Figure 1); Figure 2 details its trained components. We write $X^{\uparrow n}$ for bilinear resizing of a map X to $n \times n$ and s_k for stride k relative to the 352×352 input.

A. FROZEN ENCODER WITH CONVOLUTIONAL ADAPTERS

The encoder is DINOv3 ViT-S/16 [8], kept frozen. The benchmark gives every model 352×352 frames; DANG resizes them to 448×448 inside the network so that the 16-pixel patches form a 28×28 grid of 384-dimensional tokens. Each of the 12 blocks is followed by a residual convolutional bottleneck adapter of rank $r = 16$, in the spirit of Conv-LoRA [37] and Convpass [38], that acts on the patch tokens x of the block output:

$$x \leftarrow x + W_{\uparrow}\,\mathrm{GELU}\big(z + \mathcal{K}_{3\times3}(z)\big), \qquad z = W_{\downarrow}x, \tag{1}$$

where $W_{\downarrow} \in \mathbb{R}^{r\times384}$ and $W_{\uparrow} \in \mathbb{R}^{384\times r}$ are trained, $W_{\uparrow}$ is initialised to zero so that training starts from the frozen encoder, and $\mathcal{K}_{3\times3}$ is a depthwise convolution on the 28×28 token grid. The class and register tokens take the linear path $W_{\uparrow}W_{\downarrow}x$ only. Unlike LoRA, the adapter is a non-linear residual module, so it cannot be merged into the frozen weights; the adapters add 0.15M parameters. We read the normalised patch tokens after blocks 3, 6, 9 and 12 and project each with a 1×1 convolution, batch normalisation and activation to a map $T_\ell \in \mathbb{R}^{192\times28\times28}$, $\ell \in \{3, 6, 9, 12\}$.

B. TOKEN PYRAMID AND SCALE-AWARE DECODER

Following the token pyramid of the extended SegDINO [33], each depth is assigned to one stride, shallow depths to fine strides:

$$\begin{aligned} P_4 &= \phi_4\big(\mathcal{U}(T_3^{\uparrow44})\big), \quad & P_8 &= \phi_8\big(T_6^{\uparrow44}\big), \\ P_{16} &= \phi_{16}\big(T_9^{\uparrow22}\big), & P_{32} &= \phi_{32}\big(\mathcal{D}(T_{12}^{\uparrow22})\big), \end{aligned} \tag{2}$$

where $\mathcal{U}$ is a learned depthwise $2\times$ transposed convolution, $\mathcal{D}$ a depthwise convolution with stride 2, and ϕ_s light convolution blocks (separable 3×3 at s_4 and s_8, 1×1 at s_{16} and s_{32}). The pyramid has no convolutional backbone: all four levels come from the frozen encoder.

The scale-aware decoder of [33] refines each level with a residual depthwise block

$$R(x) = x + \gamma\,\mathrm{GELU}\big(\mathrm{GN}(W_{1\times1}\,\mathcal{K}_{3\times3}(x))\big), \tag{3}$$

with a learned scalar γ initialised to zero and group normalisation GN, and then fuses the levels top-down,

$$\begin{aligned} X_{32} &= R'_{32}\big(R_{32}(P_{32})\big), \\ X_k &= R'_k\big(X_{2k}^{\uparrow} + R_k(P_k)\big), \quad k = 16, 8, 4, \end{aligned} \tag{4}$$

where R_k and R'_k are separate blocks of the form of Eq. (3). A 1×1 convolution on X_4 gives the decoder logit Z_{SAD} at s_4 (88×88). For deep supervision during training, 1×1 heads on X_8, X_{16} and X_{32} give auxiliary logits A_8, A_{16} and A_{32}; they are not used at inference.

C. DUAL-PROTOTYPE MARGIN HEAD

The head concatenates the three deepest maps T_6, T_9 and T_{12}, projects them with a 3×3 convolution block and a 1×1 convolution to 128 channels, and normalises each position to a unit embedding $e_{u,v}$ on the 28×28 grid. It compares $e_{u,v}$ with $K = 8$ learned foreground prototypes p_k^+ and $K = 8$ background prototypes p_k^- (unit vectors, orthogonally initialised), and with a foreground anchor a^+ and background anchor a^-:

$$\begin{aligned} \pi_{\mathrm{prior}}(u,v) = \tfrac{\tau}{\beta}\Big(\log\textstyle\sum_k e^{\beta\langle e_{u,v},p_k^+\rangle} - \log\sum_k e^{\beta\langle e_{u,v},p_k^-\rangle}\Big) \\ + w_{\mathrm{a}}\big(\langle e_{u,v},a^+\rangle - \langle e_{u,v},a^-\rangle\big) + b. \end{aligned} \tag{5}$$

The log-sum-exp with fixed sharpness β is a soft maximum over prototypes; the temperature τ, the bias b and the bounded anchor weight w_{a} are learned. During training only, $a^{\pm}$ are exponential moving averages of the embeddings at positions where the ground-truth mask, average-pooled to 28×28, is mostly foreground or mostly background; they start as the normalised means of the prototypes and are fixed at test time. At training and test time the head then forms image-specific self-support centroids [35] weighted by the prior's own confidence, $c^+ = \mathrm{Norm}(\sum_{u,v} q_{u,v}^2 e_{u,v})$ and $c^- = \mathrm{Norm}(\sum_{u,v}(1-q_{u,v})^2 e_{u,v})$ with $q = \sigma(\pi_{\mathrm{prior}})$ computed from a stop-gradient copy of the prior and a floor of 10^{-4} on both weights, and outputs the margin map

$$\pi_{\mathrm{proto}} = \pi_{\mathrm{prior}} + w_{\mathrm{ss}}\big(\langle e,c^+\rangle - \langle e,c^-\rangle\big), \tag{6}$$

with a learned bounded weight w_{ss}. The final logit is

$$Z = \big(Z_{\mathrm{SAD}} + 0.2\,\pi_{\mathrm{proto}}^{\uparrow88}\big)^{\uparrow352}. \tag{7}$$

The prototype margin thus enters the output as a coarse, token-level prior on top of the decoder logit. Constant values are listed in Appendix A.

D. TRAINING OBJECTIVE

DANG is trained with

$$\begin{aligned} \mathcal{L} = \mathcal{L}^w_{\mathrm{BCE+Dice}}(Z,Y) + 0.25\,\mathcal{L}^w_{\mathrm{Tv}}(Z,Y) + 0.1\,\mathcal{L}_{\mathrm{far}}(Z,Y) \\ + \tfrac{0.25}{3}\textstyle\sum_{k\in\{8,16,32\}}\mathcal{L}^w_{\mathrm{BCE+Dice}}(A_k^{\uparrow352},Y) + 0.35\,\mathcal{L}_{\mathrm{proto}}. \end{aligned} \tag{8}$$

$\mathcal{L}_{\mathrm{BCE+Dice}}$ is 0.5 binary cross-entropy plus 0.5 soft Dice with smoothing 1. In the terms marked w, each image is weighted by w, which grows as lesion area shrinks and is normalised to mean 1 in the batch. The Tversky term $\mathcal{L}_{\mathrm{Tv}}$ (smoothing 1) penalises false negatives more for small lesions. $\mathcal{L}_{\mathrm{far}}$ is the unweighted mean over all pixels of the predicted probability times the average of four indicators of lying outside the ground truth dilated by 4, 8, 14 and 20 pixels. $\mathcal{L}_{\mathrm{proto}}$

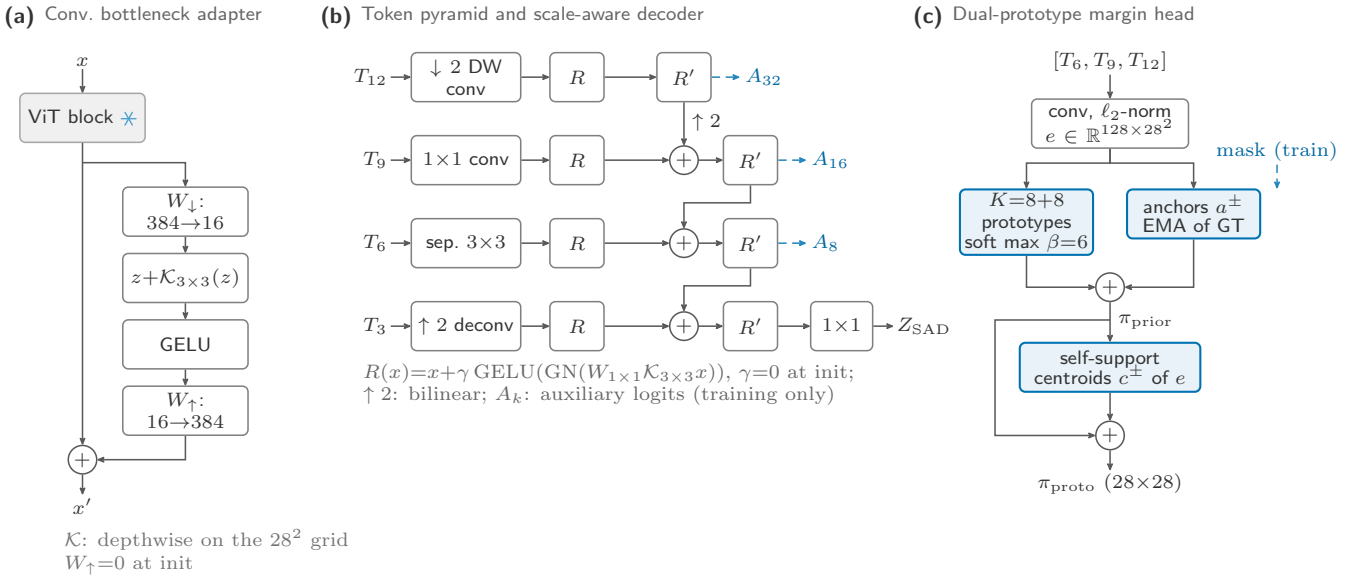


FIGURE 2. The three trained components of DANG. (a) The convolutional bottleneck adapter after each frozen ViT block (Eq. (1)): a rank-16 bottleneck with a depthwise 3 × 3 convolution on the 28 × 28 patch-token grid, whose up-projection starts at zero. (b) The token pyramid (left column, Eq. (2)); levels at 11^2, 22^2, 44^2 and 88^2) and the scale-aware decoder (Eqs. (3) and (4)): each level is refined by a residual depthwise block R, fused top-down with the upsampled coarser level and refined again by R'; auxiliary 1 × 1 heads give the deep-supervision logits A_{32}, A_{16}, A_8. (c) The dual-prototype margin head (Eqs. (5) and (6)): unit embeddings are scored against learned prototypes and ground-truth anchors, which are updated from the mask during training only, and the resulting prior is refined by image-specific self-support centroids.

applies the cross-entropy, Dice and Tversky terms to π_{proto} against the mask average-pooled to 28 × 28. Exact forms are in Appendix A. During training, each image's amplitude spectrum is mixed with that of another image in the batch with probability 0.5, keeping its phase (Fourier amplitude mix [40]), to vary colour and illumination statistics.

IV. EXPERIMENTAL SETUP

A. DATA

We use the standard split [4]: 900 Kvasir-SEG and 550 CVC-ClinicDB images for training, and the five standard test sets kept exactly as published (Table 1). Because these test sets are small, we add two multi-centre cohorts. From PolypGen [10] we keep 1,399 of 1,536 frames from its six centres, after dropping 125 frames with empty masks and 12 within-set duplicates. From PolypDB [11] we keep 1,376 frames in six groups: white light, BLI, FICE and LCI from the Vietnamese centre, white light from the Swedish centre, and NBI; we drop its Norwegian white-light group, which comes from the hospital that collected Kvasir-SEG. The kept NBI group also comes from Norwegian and Swedish sources and may include that hospital, so for this group only the imaging modality, not necessarily the centre, is new. Every kept frame is compared against all 1,450 training images and the frames already kept from its own set. Two frames count as duplicates when their MD5 hashes match, or when the Pearson correlation of their 32 × 32 grey-scale thumbnails, taken after cropping the dark endoscope border, is at least 0.97 and their masks overlap with IoU at least 0.8. A 64-bit difference hash was tried first and rejected because the black border dominates it on endoscopy frames. The highest thumbnail correlation of a kept frame with any training image is 0.94 for PolypGen and 0.92 for PolypDB.

TABLE 1. Cohorts. The five standard test sets are kept as published; PolypGen and PolypDB are deduplicated against the training images. Primary: the development cohorts our comparisons focus on. PolypDB is the held-out final test, evaluated once.

Cohort	Source	Images	Role
Train	Kvasir-SEG + CVC-ClinicDB	1,450	training
Kvasir	Kvasir-SEG [3]	100	in domain
ClinicDB	CVC-ClinicDB	62	in domain
ColonDB	CVC-ColonDB [5]	380	primary
ETIS	ETIS-LaribPolypDB [6]	196	primary
CVC-300	EndoScene test [4]	60	external
PolypGen	PolypGen, 6 centres [10]	1,399	primary
PolypDB	PolypDB, 6 groups [11]	1,376	final test

Following the standard protocol, the benchmark has no validation split, and DANG's design was developed while inspecting the test cohorts. We therefore treat CVC-ColonDB, ETIS and PolypGen, the three cohorts that are external to the training centres and large enough for per-image statistics, as primary development cohorts, and PolypDB, which no design decision used, as the confirmatory test. Kvasir and ClinicDB share their training centres, and ClinicDB and CVC-300 are small (one image can move mean Dice by up to 1.6 and 1.7 points); we report them for comparability.

B. MODELS AND TRAINING

DANG is trained for 20 epochs at 352 × 352 with batch size 16, AdamW, linear warm-up and cosine decay, gradient clipping, mixed precision and an exponential moving average

of the weights; all hyperparameters are listed in Table 9. There is no validation split and no checkpoint selection: we evaluate the moving-average weights after the last epoch.

Published scores rest on incompatible splits and are rarely tested for significance [21], and the released training scripts of some models select checkpoints on test data. We therefore retrain the compared models instead of copying their numbers. PraNet [4], Polyp-PVT [15] and PraNet-V2 [18] are retrained from their official code with the training settings of their papers and released scripts, and SegDINO [9], which uses the same frozen DINOv3 ViT-S/16 encoder as DANG, with those of its paper. All models are thus evaluated under one protocol but trained with their own recipes, and every model is evaluated at its final epoch. When a seed was run twice, the earlier run is used.

C. METRICS AND STATISTICS

All models are scored by one evaluator. Logits are resized bilinearly to the native ground-truth resolution, passed through the sigmoid and thresholded at 0.5; Dice is computed per image and averaged over the images of a cohort (for PolypGen and PolypDB, over all frames of all centres). For comparison with published tables we also report the toolbox mean Dice, the Dice averaged over 256 thresholds of the min-max-normalised probability map, which is the metric of the evaluation scripts behind PraNet, Polyp-PVT and PraNet-V2. As secondary metrics we report boundary F1 (contours matched within 2 pixels), the 95th-percentile Hausdorff distance (HD95) at 352×352, and the failure rate, the share of frames with Dice below 50%.

Five seeds (42–46) are run for every model, and the seed is the unit of inference. At the seed level, a two-sided paired t-test compares the per-seed mean Dice of DANG and each compared model over matched seed labels ($n = 5$); because runs of different codebases share no randomness, we also report Welch's unpaired test as a sensitivity check. For each compared model, Holm's procedure [42] corrects the seed-level tests over the three primary cohorts; as a second check we apply Holm jointly over all four published models and three cohorts. At the image level, a two-sided Wilcoxon signed-rank test [41] compares per-image Dice averaged over the five seeds, with zero differences dropped. Frames from the same video are not independent, and the test is conditional on the five trained models, so we use it only descriptively, together with the counts of images on which each model is better; no claim in this paper rests on image-level tests alone. Statistics were computed with SciPy. Differences quoted in the text are computed from unrounded means and can differ by 0.1 point from the difference of the rounded table entries.

Inference cost is measured for all models in one process on one NVIDIA T4 at batch size 1 and 352×352 input (256×256 for SegDINO, its own resolution): the median of 300 timed device-side forward passes after 50 warm-up passes on resident input tensors (no data transfer or post-processing), averaged over two passes of the whole benchmark, which differ by at most 0.1 ms with CUDA graphs and by up to 1.0 ms in the other two modes. We time three modes: FP16 autocast (eager), FP16 weights, and FP16 weights replayed from a CUDA graph. We also report multiply-accumulate operations (MACs) per frame and the parameters of the timed model, frozen weights included.

V. RESULTS

A. COMPARISON WITH RETRAINED PUBLISHED MODELS

Table 2 compares DANG with the four retrained models.

a: Held-out test.

On PolypDB, evaluated once after all design decisions, DANG reaches $89.8 \pm 0.1\%$: 3.1 points above Polyp-PVT (86.7%; 95% confidence interval of the seed-level difference 1.5 to 4.7 points, paired $p = 0.005$, Welch $p = 0.005$), 4.4 and 5.0 points above PraNet-V2 and PraNet, and 17.4 points above SegDINO. Averaged over the six PolypDB groups instead of over frames, DANG reaches 88.3% against 84.3% for Polyp-PVT. Most frames are segmented about equally well by the two models: DANG is better on 713 frames and worse on 662, and the median per-frame difference is below 0.1 point, so the higher mean comes mainly from fewer failed frames (3.8 against 7.5% of frames below 50% Dice, Table 4).

b: Development cohorts.

Among the retrained models, Polyp-PVT is the strongest on ColonDB (80.9%) and PolypGen (81.4%), and PraNet-V2 on ETIS (75.4%). DANG reaches 80.7% on ColonDB, 82.7% on ETIS and 84.2% on PolypGen. On ETIS and PolypGen it is 7.3 and 2.7 points above the best retrained model, with the smallest seed-to-seed SD of the compared models on the three primary cohorts (0.2–0.7 points). On ColonDB its difference to Polyp-PVT, -0.2 points (95% confidence interval -1.4 to $+1.1$), is not significant.

c: In domain and CVC-300.

On Kvasir DANG has the highest mean (92.2%, 1.2 points above Polyp-PVT), but it is better than Polyp-PVT on only 40 of the 100 images; the higher mean comes from a few frames on which Polyp-PVT fails. On ClinicDB it is 3.2 points below Polyp-PVT and worse on 55 of the 62 images, a systematic deficit; a plausible reason, which we did not test, is that the fully fine-tuned encoders of the compared models fit the training centres more closely than a frozen encoder. On CVC-300 its mean is 0.9 points below PraNet, although it is better on 40 of the 60 images.

d: SegDINO.

SegDINO, which uses the same frozen encoder with a light decoder, stays far below every other model, at 51.5% on ColonDB and 43.3% on ETIS. It is trained at the 256×256 input of its paper and without adapters, so this gap mixes input resolution, adaptation and decoder; Section V-B shows that DANG's token pyramid and decoder on the fully frozen encoder still exceed it by 14.1–28.7 points. We keep it as the

TABLE 2. Comparison with retrained published models. Dice (%) at threshold 0.5, mean ± SD over 5 seeds, final-epoch weights, one evaluator. ColonDB, ETIS and PolypGen are the primary cohorts; PolypDB is the held-out final test, evaluated once for the selected model. Bold: best in column. *: worse than DANG after Holm correction (seed-level paired *t*-test over the primary cohorts, Section IV-C); PolypDB tests are reported in the text. PraNet's 32.5M include an unused 2.05M classification layer of the official Res2Net-50. Params: total parameters in millions, frozen weights included; Trained: parameters not frozen in training.

			In domain		Primary (external)			External	Final test
Model	Params	Trained	Kvasir	ClinicDB	ColonDB	ETIS	PolypGen	CVC-300	PolypDB
PraNet [4]	32.5	32.5	$90.0_{\pm 0.8}$	$90.9_{\pm 1.1}$	$71.9_{\pm 2.3}$*	$65.0_{\pm 2.2}$*	$79.0_{\pm 0.7}$*	$\mathbf{89.8}_{\pm 0.6}$	$84.8_{\pm 0.7}$
Polyp-PVT [15]	25.1	25.1	$91.0_{\pm 0.7}$	$\mathbf{92.9}_{\pm 0.7}$	$\mathbf{80.9}_{\pm 0.8}$	$74.6_{\pm 2.4}$*	$81.4_{\pm 1.3}$*	$88.2_{\pm 1.0}$	$86.7_{\pm 1.3}$
PraNet-V2 [18]	30.8	30.8	$90.7_{\pm 0.7}$	$91.9_{\pm 1.2}$	$76.2_{\pm 1.6}$*	$75.4_{\pm 0.9}$*	$80.7_{\pm 0.7}$*	$88.4_{\pm 1.3}$	$85.4_{\pm 1.4}$
SegDINO [9]	24.1	2.48	$81.6_{\pm 0.9}$	$74.1_{\pm 2.1}$	$51.5_{\pm 4.6}$*	$43.3_{\pm 2.9}$*	$64.2_{\pm 1.1}$*	$70.5_{\pm 1.2}$	$72.4_{\pm 0.7}$
DANG (ours)	23.5	1.94	$\mathbf{92.2}_{\pm 0.4}$	$89.7_{\pm 1.3}$	$80.7_{\pm 0.3}$	$\mathbf{82.7}_{\pm 0.7}$	$\mathbf{84.2}_{\pm 0.2}$	$88.8_{\pm 1.9}$	$\mathbf{89.8}_{\pm 0.1}$

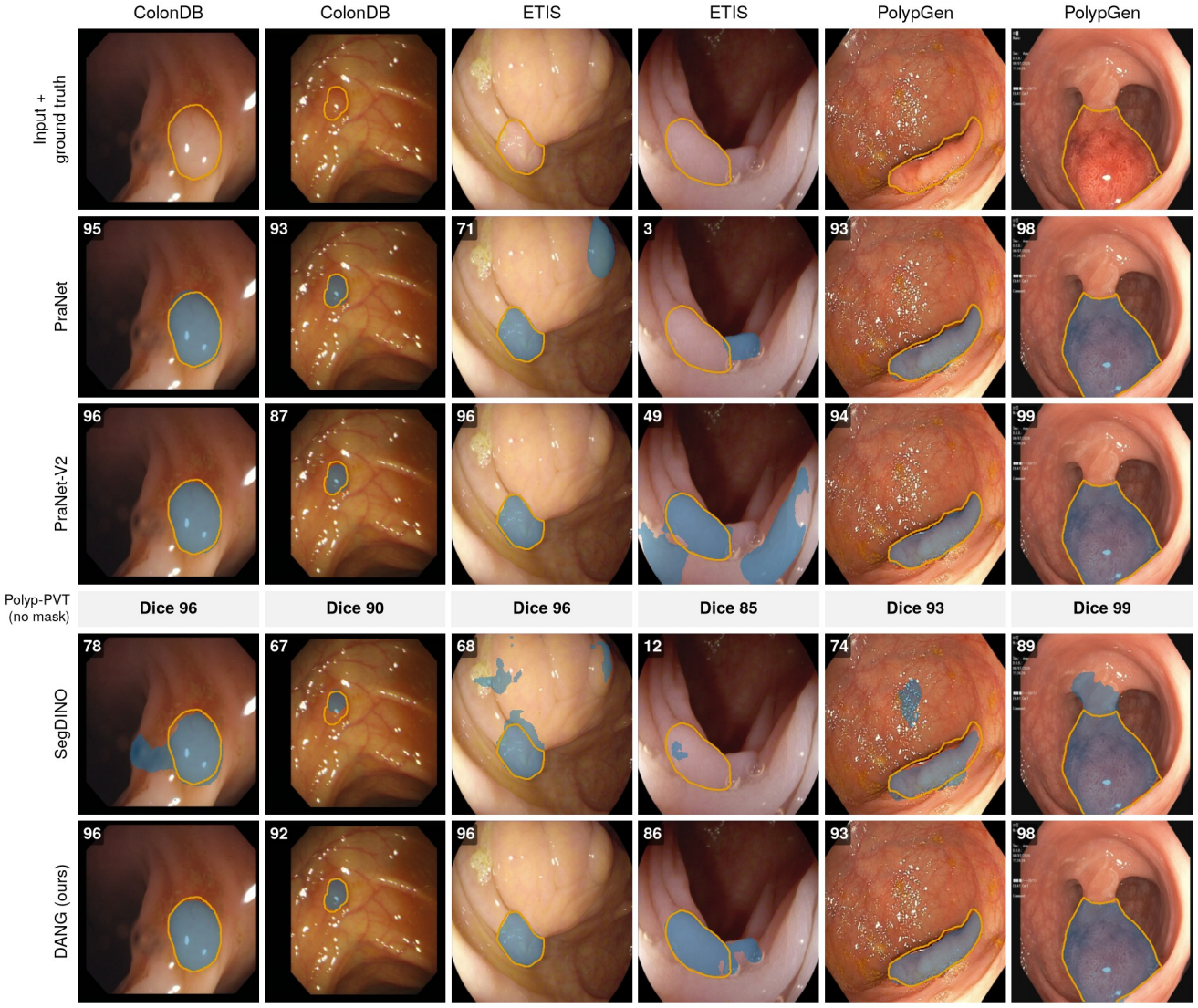


FIGURE 3. Predictions on the primary cohorts, drawn at random. Two frames per cohort, chosen by a fixed random generator among all frames (not curated); seed 42 of each model. Orange contour: ground truth; blue fill: prediction at threshold 0.5, shown at the network's output resolution; number: per-image Dice (%) at native resolution. Polyp-PVT's prediction masks were not retained; its strip gives the per-image Dice it logged for the same frames and seed.

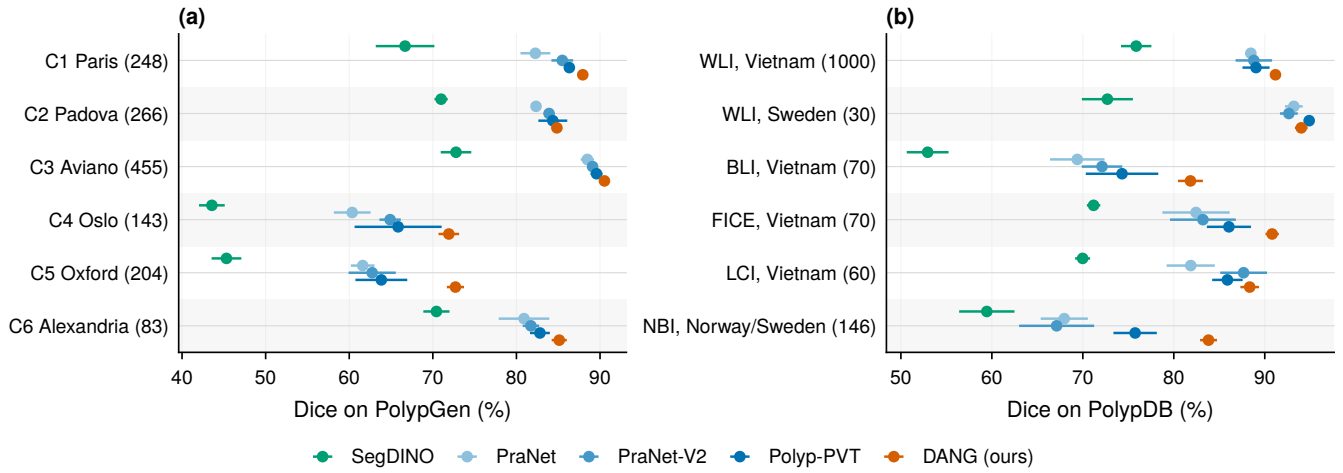


FIGURE 4. DANG gains most on the centres where every model is weakest. Dice (%) per centre of PolypGen (a) and per centre and imaging modality of PolypDB (b); number of frames in parentheses. Points are means over 5 seeds; bars are 95% *t*-intervals over training seeds, not over patients or centres.

reference for a frozen foundation model with a light head: DANG uses the same frozen weights and trains a comparable number of parameters (1.9M against 2.5M).

e: Per centre.

Figure 4a splits PolypGen by centre. DANG has the highest mean Dice on all six centres. Its margin over the best retrained model is 0.5–2.3 points on Paris, Padova, Aviano and Alexandria and 6.1 and 8.9 points on Oslo and Oxford, the two centres on which every retrained model stays below 66%. On PolypDB (Figure 4b), DANG is the best model on five of the six groups. Its gains are largest on narrow-band imaging (83.8 against 75.8% for Polyp-PVT) and blue-light imaging (81.8 against 74.3%), two of the four virtual-chromoendoscopy modalities absent from training; on the 30 Swedish white-light frames it is 0.9 points below Polyp-PVT. These per-group values are five-seed means without per-group tests, and four of the six groups have 70 frames or fewer. Averaged over centres, PolypGen gives 82.2% for DANG against 78.8% for Polyp-PVT.

f: Examples.

Figure 3 shows six frames drawn at random from the primary cohorts. Four are segmented well by PraNet, PraNet-V2, Polyp-PVT and DANG; on the first ETIS frame PraNet adds a false-positive region (Dice 71% against 96% for the other three). SegDINO adds scattered false positives on several of these five frames (Dice 67–89%). On the second ETIS frame PraNet finds only a fragment of the lesion (Dice 3%), PraNet-V2 adds large false positives on the surrounding folds (49%) and SegDINO reaches 12%, while Polyp-PVT and DANG reach 85 and 86%.

g: Statistical tests.

Table 3 gives the tests on the primary cohorts. On ETIS and PolypGen, DANG is better than every retrained model at the seed level after Holm correction; against Polyp-PVT the adjusted p-values are 0.007 and 0.031. The conclusions are unchanged with Welch's unpaired test ($p = 0.001$ and 0.008 against Polyp-PVT) and with one Holm family over all twelve comparisons with published models (largest adjusted p for these claims 0.031). Descriptively, DANG is better on 123 of 196 ETIS images (worse on 70) and on 746 of 1,399 PolypGen frames (worse on 647). On ColonDB, DANG is better than PraNet, PraNet-V2 and SegDINO and not significantly different from Polyp-PVT (adjusted $p = 0.71$; 187 images better, 191 worse).

h: Boundaries and failures.

Table 4 reports secondary metrics on the three primary cohorts and PolypDB. DANG has the lowest HD95 and the lowest failure rate on all four; on ETIS, for example, 7.2% of its frames fall below 50% Dice, against 17.2–51.9% for the retrained models. Its boundary F1, however, is 1.5–3.5 points below that of Polyp-PVT on ColonDB, PolypGen and PolypDB and higher only on ETIS. DANG's Dice gains therefore come mainly from fewer missed or fragmented lesions rather than from sharper contours; the stride-16 token grid of the encoder may limit boundary precision.

i: Link to published numbers.

Table 5 reports the toolbox metric next to published scores. Our Polyp-PVT is within 1.2 points of its published scores on Kvasir, ClinicDB and ColonDB, and 2.2 and 4.3 points below them on CVC-300 and ETIS. ETIS has the largest seed-to-seed spread of the five sets, and part of the published advantage may come from selecting the checkpoint on test data. Our PraNet is 0.3 points below its published score on Kvasir and above it on the other four sets. Under this metric DANG leads on ETIS (80.9%, also above the published 78.7% of Polyp-PVT), ties Polyp-PVT on Kvasir and CVC-300, and ranks below it on ClinicDB and ColonDB (79.4 against 80.6%).

TABLE 3. Paired tests of DANG against each retrained model on the primary cohorts. Δ: mean Dice difference (pp, DANG minus model) over 5 matched seeds; p_{seed}: paired t-test over seeds; p_{img}: Wilcoxon signed-rank test over images on seed-averaged Dice, with the number of images on which DANG is better/worse; descriptive only, because frames from one video are not independent. All p-values Holm-corrected over the three cohorts.

	ColonDB ($n = 380$)			ETIS ($n = 196$)			PolypGen ($n = 1{,}399$)		
	Δ	p_{seed}	p_{img}	Δ	p_{seed}	p_{img}	Δ	p_{seed}	p_{img}
PraNet	+8.7	0.001	<0.001 (255/122)	+17.7	<0.001	<0.001 (140/51)	+5.2	<0.001	<0.001 (844/548)
Polyp-PVT	−0.2	0.715	0.989 (187/191)	+8.1	0.007	<0.001 (123/70)	+2.7	0.031	<0.001 (746/647)
PraNet-V2	+4.5	0.004	<0.001 (233/144)	+7.3	<0.001	<0.001 (123/70)	+3.4	<0.001	<0.001 (765/630)
SegDINO	+29.2	<0.001	<0.001 (365/12)	+39.5	<0.001	<0.001 (187/4)	+19.9	<0.001	<0.001 (1301/93)

TABLE 4. Secondary metrics on the primary cohorts and PolypDB. Boundary F1 (BF1, %, contours within 2 pixels; higher is better), HD95 (pixels at 352 × 352; lower is better) and failure rate (% of frames with Dice below 50%; lower is better); means over 5 seeds. **Bold**: best in column.

	BF1 ↑				HD95 ↓				Failure rate ↓			
Model	ColonDB	ETIS	PolypGen	PolypDB	ColonDB	ETIS	PolypGen	PolypDB	ColonDB	ETIS	PolypGen	PolypDB
PraNet	42.9	46.5	54.7	64.5	45.4	57.5	43.7	27.3	20.6	27.4	14.2	8.8
Polyp-PVT	**49.0**	53.6	**58.7**	**68.9**	38.9	53.3	45.3	30.0	10.9	17.6	12.2	7.5
PraNet-V2	45.7	53.5	58.0	67.2	47.3	49.4	45.8	31.5	16.6	17.2	12.6	8.6
SegDINO	20.9	19.4	28.8	33.1	52.3	69.4	55.4	41.5	40.6	51.9	27.9	16.3
DANG (ours)	47.5	**56.2**	55.2	66.8	**34.7**	**31.0**	**33.6**	**19.1**	**10.6**	**7.2**	**8.1**	**3.8**

TABLE 5. Toolbox mean Dice on the standard test sets (%), the metric of published polyp tables. Published rows are copied from the papers; ‡: the released training script selects its checkpoint on test data. Our rows: mean ± SD over 5 seeds, final-epoch weights. **Bold**: best of our rows.

Model	Kvasir	ClinicDB	ColonDB	ETIS	CVC-300
PraNet (published) [4]	89.8	89.9	70.9	62.8	87.1
Polyp-PVT (published)‡ [15]	91.7	93.7	80.8	78.7	90.0
PraNet [4]	$89.5_{\pm 0.7}$	$90.5_{\pm 1.1}$	$72.1_{\pm 2.3}$	$65.8_{\pm 2.0}$	$\mathbf{89.2}_{\pm 0.6}$
Polyp-PVT [15]	$\mathbf{90.7}_{\pm 0.7}$	$\mathbf{92.5}_{\pm 0.7}$	$\mathbf{80.6}_{\pm 0.8}$	$74.4_{\pm 2.4}$	$87.8_{\pm 1.0}$
PraNet-V2 [18]	$90.4_{\pm 0.7}$	$91.5_{\pm 1.2}$	$75.9_{\pm 1.5}$	$75.2_{\pm 0.9}$	$88.1_{\pm 1.3}$
SegDINO [9]	$77.9_{\pm 0.9}$	$70.9_{\pm 1.2}$	$51.8_{\pm 2.7}$	$44.2_{\pm 1.4}$	$65.6_{\pm 0.5}$
DANG (ours)	$\mathbf{90.7}_{\pm 0.4}$	$88.4_{\pm 1.1}$	$79.4_{\pm 0.3}$	$\mathbf{80.9}_{\pm 0.6}$	$87.8_{\pm 1.8}$

B. COMPONENT ABLATION

Table 6 removes each trained component of DANG in turn and then both, over five seeds; the last row (frozen encoder, token pyramid and decoder) is the closest baseline.

a: Adapters.

Without the adapters, the fully frozen encoder costs 7.3, 10.8 and 5.5 points on ColonDB, ETIS and PolypGen (seed-level adjusted $p < 0.001$), the largest effect in the paper: the 0.15M adapter parameters are what lets the frozen encoder transfer to colonoscopy. Without adapters or prototype head, the token pyramid and scale-aware decoder still exceed SegDINO by 21.5, 28.7 and 14.1 points, a gap that mixes decoder, input resolution (448 against 256) and recipe; SegDINO's low scores (Table 2) are thus not due to the missing adapters.

b: Prototype head.

Removing the dual-prototype margin head, its logit term and its loss lowers Dice by 0.7, 1.0 and 0.2 points on the primary cohorts. The effect is small and not significant over five seeds (adjusted $p = 0.20$, 0.31 and 0.31; the 95% confidence intervals of the differences, −0.1 to 1.4, −0.6 to 2.5 and −0.1 to 0.6 points, include zero). It is slightly negative on Kvasir and CVC-300 (−0.2, −0.4), and similar on the frozen encoder (+0.4, 0.0, +0.4; last two rows). We keep the head because its effect points the same way on all three primary cohorts, but the evidence for it is weak.

C. CHOICE OF DECODER

Table 7 compares three decoder configurations, each with its own deep supervision, on the same frozen DINOv3 ViT-S/16 with adapters and the same token pyramid, with the same loss family, recipe, evaluator and T4 hardware, over five seeds each. Besides the scale-aware decoder of DANG, we test the fg/bg decoder with PASSE described in the caption, and a query decoder in which one foreground and one background query attend to the pyramid with masked cross-attention.

a: Accuracy and cost.

Over five seeds the fg/bg decoder with PASSE is the most accurate variant: it is 1.0, 0.4 and 0.5 points above DANG on ColonDB, ETIS and PolypGen. The differences are small and not significant at the seed level after Holm correction (adjusted $p = 0.064$, 0.116 and 0.076); descriptively, the fg/bg decoder is better on 251 ColonDB images and worse on 128. The query decoder lies between the two on ColonDB and PolypGen and below both on ETIS; none of its differences to DANG is significant at the seed level, and descriptively it is better than DANG on 217 of 380 ColonDB images and 886 of 1,399 PolypGen frames. We select the scale-aware decoder for its cost: against the fg/bg decoder with PASSE it removes 1.5M parameters and 31% of the MACs (25.1G against 36.1G) and cuts the T4 latency by 19% (9.4 ms against 11.6 ms), for a loss of at most 1.0 point on the

TABLE 6. Component ablation of DANG. Dice (%) at threshold 0.5, mean ± SD over 5 seeds, same recipe, evaluator and T4 hardware. Δ: mean Dice difference on the primary cohorts, DANG minus the variant (pp). *: significant at the seed level after Holm correction over the three cohorts (paired *t*-test). Params and Trained in millions.

Variant	Params	Trained	Kvasir	ClinicDB	ColonDB	ETIS	PolypGen	CVC-300	Δ (DANG − variant) ColonDB	ETIS	PolypGen
DANG	23.5	1.94	$92.2_{\pm 0.4}$	$89.7_{\pm 1.3}$	$80.7_{\pm 0.3}$	$82.7_{\pm 0.7}$	$84.2_{\pm 0.2}$	$88.8_{\pm 1.9}$	–	–	–
without prototype head	22.5	0.92	$92.4_{\pm 0.2}$	$89.5_{\pm 0.9}$	$80.0_{\pm 0.6}$	$81.8_{\pm 0.7}$	$84.0_{\pm 0.2}$	$89.2_{\pm 1.0}$	+0.7	+1.0	+0.2
without adapters (frozen encoder)	23.4	1.79	$88.5_{\pm 0.2}$	$87.2_{\pm 0.9}$	$73.4_{\pm 0.5}$	$71.9_{\pm 1.1}$	$78.7_{\pm 0.3}$	$86.4_{\pm 1.0}$	+7.3*	+10.8*	+5.5*
without adapters and prototype head	22.4	0.77	$88.3_{\pm 0.2}$	$86.6_{\pm 0.7}$	$73.0_{\pm 0.3}$	$71.9_{\pm 0.5}$	$78.3_{\pm 0.3}$	$85.8_{\pm 0.7}$	+7.7*	+10.8*	+5.8*

TABLE 7. Decoder comparison on the same frozen encoder and token pyramid. Dice (%) at threshold 0.5, mean ± SD over 5 seeds; all models use the frozen DINOv3 ViT-S/16 with convolutional adapters, the recipe of Section IV-B and T4 hardware. fg/bg decoder: four top-down stages predicting foreground and background logits gated by boundary uncertainty, with a 768-channel fusion head; PASSE: prototype-anchored spatial support envelope on the output logits. Params in millions; GMACs per frame; latency in ms at batch 1, FP16 CUDA graph on an NVIDIA T4. **Bold**: best in column.

Decoder	Params	GMACs	Latency	Kvasir	ClinicDB	ColonDB	ETIS	PolypGen	CVC-300
fg/bg decoder + PASSE	25.0	36.1	11.6	$92.2_{\pm 0.4}$	$90.6_{\pm 1.2}$	$\mathbf{81.6}_{\pm 0.6}$	$\mathbf{83.1}_{\pm 0.4}$	$\mathbf{84.6}_{\pm 0.2}$	$\mathbf{89.2}_{\pm 1.0}$
fg/bg query decoder	25.4	25.5	10.1	$\mathbf{92.3}_{\pm 0.6}$	$\mathbf{90.9}_{\pm 0.5}$	$81.3_{\pm 0.8}$	$82.3_{\pm 1.4}$	$84.4_{\pm 0.3}$	$88.8_{\pm 1.0}$
DANG: scale-aware decoder	23.5	25.1	9.4	$92.2_{\pm 0.4}$	$89.7_{\pm 1.3}$	$80.7_{\pm 0.3}$	$82.7_{\pm 0.7}$	$84.2_{\pm 0.2}$	$88.8_{\pm 1.9}$

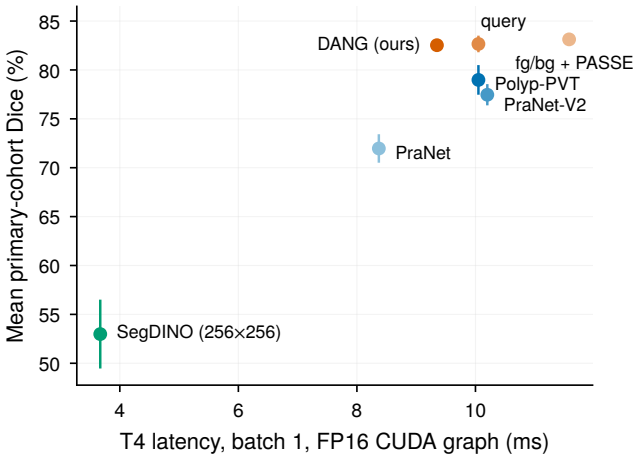


FIGURE 5. DANG is more accurate than the retrained models at a comparable latency. Mean Dice over the three primary cohorts against batch-1 latency on one NVIDIA T4 (FP16 weights, CUDA graph; for DANG, FP16 weights change Dice by less than 0.03 points). Points are means over 5 seeds, bars 95% *t*-intervals of the per-seed mean. Blue: retrained published models; green: SegDINO, timed at its 256 × 256 input; orange: decoders on the DANG encoder and token pyramid.

primary cohorts. Where accuracy matters more than latency, the fg/bg decoder with PASSE is the better choice. Because these variants were compared on the development cohorts, this choice is itself part of the adaptive design discussed in Section VI.

D. EFFICIENCY

Table 8 lists the cost of each model, and Figure 5 plots accuracy against latency. DANG has the fewest total and trained parameters of all compared models, but about twice the MACs per frame of the Res2Net and PVTv2 models (25.1G against 10.8–13.0G), because the frozen encoder runs at 448×448. With FP16 weights in a CUDA graph its batch-1 latency (9.4 ms, about 107 forward passes per second on the device) is below that of Polyp-PVT (10.1 ms) and PraNet-V2 (10.2 ms) and above that of PraNet (8.4 ms); the end-to-end time of a clinical pipeline would add transfer and post-processing. This advantage depends on CUDA graphs: with FP16 weights alone, Polyp-PVT (18.3 ms) is faster than DANG (20.7 ms). FP16 weights do not change DANG's accuracy: in the training runs, the FP16 copy of every seed scores within 0.03 points of the FP32 model on the three primary cohorts. SegDINO, at its 256×256 input, is the fastest model (3.7 ms) but far less accurate. In eager mode DANG has the lowest batch-1 latency of the 352×352 models (22.7 ms) but the lowest batch-8 throughput (41 against 155–174 images per second), because its encoder stays in FP32 there; batched offline processing would need the FP16 encoder. Training DANG takes 27 minutes per seed on one T4, against 66 and 203 minutes for PraNet and PraNet-V2 on the same GPU.

TABLE 8. Inference and training cost. Inference on one NVIDIA T4 at batch size 1 and 352 × 352 input, all models timed in one process; Params (of the timed model) and Trained in millions; MACs per frame in billions. Eager: FP16 autocast (DANG's encoder stays in FP32 in this mode, as in training); FP16: FP16 weights; graph: FP16 weights replayed from a CUDA graph; img/s: batch size 8, eager. Train: mean training time per seed in minutes on one T4. ‖: the timed PraNet omits the unused 2.05M Res2Net classifier counted in Table 2. ¶: trained on a faster cloud GPU, so its training time is not comparable. §: timed at its own 256 × 256 input.

Model	Params	Trained	GMACs	Latency (ms) eager	FP16	graph	img/s	Train
PraNet‖	30.5	30.5	13.0	26.6	25.8	8.4	174	66
Polyp-PVT¶	25.1	25.1	10.8	23.7	18.3	10.1	155	69
PraNet-V2	30.8	30.8	12.1	31.0	25.6	10.2	159	203
SegDINO§	24.1	2.48	6.9	14.6	10.9	3.7	397	16
DANG (ours)	23.5	1.94	25.1	22.7	20.7	9.4	41	27

VI. LIMITATIONS

1) **Development on the test cohorts.** Following the standard benchmark, which has no validation split, DANG's design, including the choice of decoder, was developed while inspecting the test cohorts. Results on Kvasir, ClinicDB, ColonDB, ETIS, CVC-300 and PolypGen are therefore development results, and their tests do not account for this selection. PolypDB was never used for design and was evaluated once, on the final model; it is the confirmatory result, and there DANG leads every retrained model, by 3.1 to 17.4 points.
2) **Dependent frames.** The cohorts consist of video frames, and patient or sequence identifiers are not available for all of them, so image-level tests are descriptive; our inference rests on five training seeds, which gives only four degrees of freedom.
3) **Boundary precision.** DANG's boundary F1 is below that of Polyp-PVT on ColonDB, PolypGen and PolypDB, and it trails Polyp-PVT on the in-domain ClinicDB.
4) **Recipes differ.** The compared models use the recipes of their papers, DANG its own; in particular only DANG trains with Fourier amplitude mixing and mixed precision. The comparison is between methods as published, as is usual on this benchmark; isolating the contribution of each recipe element under matched training budgets is left to future work. We also did not ablate amplitude mixing within DANG's own recipe, so part of its advantage on unseen centres may come from this augmentation rather than from the architecture.
5) **Hardware.** DANG and most compared models were trained on NVIDIA T4 GPUs; Polyp-PVT, whose 99-epoch schedule exceeds our T4 session limit, was trained on cloud GPUs of another type.
6) **One training split.** All models are trained on the same two centres; we do not test other training sources or label budgets.
7) **Scope.** We evaluate single frames, not video, and report no clinical reader study.

VII. CONCLUSION

DANG segments polyps from a frozen DINOv3 ViT-S/16 with convolutional bottleneck adapters, a token pyramid, a scale-aware decoder and a dual-prototype margin head, training 1.9M of 23.5M parameters. On the held-out PolypDB, evaluated once, it reaches 89.8% mean Dice, 3.1 points above the strongest retrained model. On the development cohorts it exceeds every compared model on ETIS-LaribPolypDB (+7.3 points) and PolypGen (+2.7 points), shows no significant difference to Polyp-PVT on CVC-ColonDB, and trails Polyp-PVT on the in-domain CVC-ClinicDB and in boundary F1; its gains come mainly from fewer failed frames. The adapters are essential (5.5–10.8 points on the primary cohorts), the prototype head adds a small 0.2–1.0 points that are not significant across seeds, and a heavier fg/bg decoder with a support envelope is up to 1 point more accurate while DANG's decoder has 19% lower latency (9.4 ms per frame on a T4 with CUDA graphs). Next steps are a validation split for design choices, training on other centres, sequence-level statistics, video-level evaluation and a reader study.

APPENDIX A IMPLEMENTATION DETAILS

DATA AND CODE AVAILABILITY

All experiments use public, de-identified datasets (Kvasir-SEG [3], CVC-ClinicDB, CVC-ColonDB [5], ETIS-LaribPolypDB [6], CVC-300, PolypGen [10] and PolypDB [11]) under their research licences; no new patient data were collected. Code, deduplication manifests, per-image evaluation logs and the scripts that regenerate every table from the raw logs will be released.

ACKNOWLEDGMENT

The authors would like to express their sincere appreciation to the Vietnam Aerospace University for supporting this research during the 2026–2027 academic year.

REFERENCES

[1] H. Sung, J. Ferlay, R. L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, and F. Bray, "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA: Cancer J. Clin.*, vol. 71, no. 3, pp. 209–249, 2021.
[2] A. G. Zauber *et al.*, "Colonoscopic polypectomy and long-term prevention of colorectal-cancer deaths," *N. Engl. J. Med.*, vol. 366, no. 8, pp. 687–696, 2012.
[3] D. Jha *et al.*, "Kvasir-SEG: A segmented polyp dataset," in *Proc. Int. Conf. MultiMedia Modeling (MMM)*, 2020, pp. 451–462.
[4] D.-P. Fan, G.-P. Ji, T. Zhou, G. Chen, H. Fu, J. Shen, and L. Shao, "PraNet: Parallel reverse attention network for polyp segmentation," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2020, pp. 263–273.
[5] N. Tajbakhsh, S. R. Gurudu, and J. Liang, "Automated polyp detection in colonoscopy videos using shape and context information," *IEEE Trans. Med. Imag.*, vol. 35, no. 2, pp. 630–644, Feb. 2016.
[6] J. Silva, A. Histace, O. Romain, X. Dray, and B. Granado, "Toward embedded detection of polyps in WCE images for early diagnosis of colorectal cancer," *Int. J. Comput. Assist. Radiol. Surg.*, vol. 9, no. 2, pp. 283–293, 2014.
[7] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.
[8] O. Siméoni *et al.*, "DINOv3," 2025, *arXiv:2508.10104*.
[9] S. Yang, H. Wang, Z. Xing, S. Chen, and L. Zhu, "SegDINO: An efficient design for medical and natural image segmentation with DINO-V3," 2025, *arXiv:2509.00833*.
[10] S. Ali *et al.*, "A multi-centre polyp detection and segmentation dataset for generalisability assessment," *Sci. Data*, vol. 10, no. 1, Art. no. 75, 2023.
[11] D. Jha *et al.*, "PolypDB: A curated multi-center dataset for development of AI algorithms in colonoscopy," 2024, *arXiv:2409.00045*.
[12] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2015, pp. 234–241.
[13] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, "UNet++: A nested U-Net architecture for medical image segmentation," in *Proc. Deep Learn. Med. Image Anal. (DLMIA) Workshop*, 2018, pp. 3–11.
[14] A. Lou, S. Guan, H. Ko, and M. H. Loew, "CaraNet: Context axial reverse attention network for segmentation of small medical objects," in *Proc. SPIE Med. Imag.*, 2022.
[15] B. Dong, W. Wang, D.-P. Fan, J. Li, H. Fu, and L. Shao, "Polyp-PVT: Polyp segmentation with pyramid vision transformers," *CAAI Artif. Intell. Res.*, vol. 2, Art. no. 9150015, 2023, doi: 10.26599/AIR.2023.9150015.
[16] Y. Zhang, H. Liu, and Q. Hu, "TransFuse: Fusing transformers and CNNs for medical image segmentation," in *Proc. Med. Image Comput. Comput. Assist. Interv. (MICCAI)*, 2021, pp. 14–24.

TABLE 9. DANG hyperparameters. Lesion area a is the ground-truth foreground percentage of the image.

Item	Setting
Encoder	DINOv3 ViT-S/16 (LVD-1689M), frozen, input 448×448 (resized from 352×352), run in FP32 during training (its attention logits overflow FP16 under autocast); FP16 weights at inference change Dice by less than 0.03 points
Adapters	residual convolutional bottleneck (Eq. (1)), rank 16, after each of the 12 blocks; readout after blocks 3, 6, 9, 12; 192 decoder channels
Optimiser	AdamW; learning rate 10^{-3} (adapters), 4×10^{-4} (all other trained modules)
Weight decay	10^{-4}; none on adapters, normalisation, bias, gain and prototype parameters
Schedule	linear warm-up for 2 epochs, then cosine decay to 5%; 20 epochs; batch size 16
Weight averaging	EMA with decay $\min(0.992,\ (1+t)/(10+t))$ at step t; final EMA weights are evaluated
Other	gradient-norm clipping at 3.0; FP16 mixed precision on T4
Augmentation	scale 0.65–0.90 ($p = 0.3$) or 0.95–1.20 with crop or pad; horizontal and vertical flips ($p = 0.5$ each); rotation by $90°$, $180°$ or $270°$ ($p = 0.5$); brightness, contrast, saturation $\pm 15\%$ ($p = 0.4$); hue ± 0.04 ($p = 0.3$); gamma 0.8–1.25 ($p = 0.3$); Fourier amplitude mix with a random batch partner, $\lambda \sim U(0,1)$ ($p = 0.5$)
Image weight	$w = \mathrm{clip}\big((3/(a+0.35))^{0.4},\ 1,\ 2.4\big)$, normalised to mean 1 in the batch
BCE+Dice	0.5 binary cross-entropy (pixel mean) $+ 0.5\,(1 -$ soft Dice), Dice smoothing 1
Tversky	false-positive weight $0.45 + 0.15\,\sigma((a-1.5)/0.5)$, false-negative weight 1 minus it, smoothing 1
Far-field loss	mean over all pixels of $\sigma(Z) \cdot \frac{1}{4}\sum_r \mathbb{1}$[outside the mask dilated by r], $r \in \{4, 8, 14, 20\}$ pixels; not image-weighted
Prototype head	$K = 8 + 8$ prototypes, dimension 128; $\beta = 6$; τ learned, initialised at 12 and clamped to $[4, 25]$ $w_{\mathrm{a}} = 10 \cdot \frac{1}{2}\tanh(\theta_{\mathrm{a}})$, $w_{\mathrm{ss}} = 8 \cdot \frac{1}{2}\tanh(\theta_{\mathrm{ss}})$ with learned θ
Anchors	momentum 0.95; foreground where pooled mask ≥ 0.35, background where ≤ 0.10; initialised to the normalised prototype means
Self-support	weights $q^2 + 10^{-4}$ and $(1-q)^2 + 10^{-4}$, q from a stop-gradient copy of π_{prior}

[17] M. M. Rahman, M. Munir, and R. Marculescu, "EMCAD: Efficient multi-scale convolutional attention decoding for medical image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024.

[18] B.-C. Hu, G.-P. Ji, D. Shao, and D.-P. Fan, "PraNet-V2: Dual-supervised reverse attention for medical image segmentation," *Comput. Vis. Media*, 2026, doi: 10.26599/CVM.2025.9450510.

[19] Y. Sun *et al.*, "ASGNet: Adaptive spectrum guidance network for automatic polyp segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, 2026, doi: 10.1109/TCSVT.2026.3685209.

[20] Y. Wang, J. Wu, J. Song, and S. Ren, "CSG-Mamba: A convolutional scoring gating vision state space network for endoscopic polyp segmentation," 2026, *arXiv:2608.14146*.

[21] A. Urooj, Z. U. Abdien, and N. Madan, "Metrics or mirage? An audit of evaluation inconsistencies in colonoscopy polyp segmentation benchmarks," 2026, *arXiv:2607.08203*.

[22] H. Xi *et al.*, "Frequency prior guided matching: A data augmentation approach for generalizable semi-supervised polyp segmentation," 2025, *arXiv:2508.06517*.

[23] T. Zhou, Y. Zhang, Y. Zhou, Y. Wu, and C. Gong, "Can SAM segment polyps?" 2023, *arXiv:2304.07583*.

[24] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, "Segment anything in medical images," *Nature Commun.*, vol. 15, Art. no. 654, 2024.

[25] C. Jiang *et al.*, "Medical SAM3: A foundation model for universal prompt-driven medical image segmentation," 2026, *arXiv:2601.10880*.

[26] T. Chen *et al.*, "SAM3-Adapter: Efficient adaptation of Segment Anything 3 for camouflage object segmentation, shadow detection, and medical image segmentation," 2025, *arXiv:2511.19425*.

[27] H. L. Li and A. Masood, "HAT-SAM3: Endoscopy-aware adaptation of a foundation segmentation model for generalizable polyp segmentation," *Vis. Comput.*, 2026, doi: 10.1007/s00371-026-04562-3.

[28] T. K. Dutta, S. Majhi, D. R. Nayak, and D. Jha, "SAM-Mamba: Mamba guided SAM architecture for generalized zero-shot polyp segmentation," 2024, *arXiv:2412.08482*.

[29] T. K. Dutta, S. Majhi, D. R. Nayak, and D. Jha, "Mamba guided boundary prior matters: A new perspective for generalized polyp segmentation," 2025, *arXiv:2507.01509*.

[30] S. Agnihotri, S. Majhi, and D. R. Nayak, "Sharpening lightweight models for generalized polyp segmentation: A boundary guided distillation from foundation models," 2026, *arXiv:2604.17865*.

[31] S. Agnihotri, S. Majhi, D. R. Nayak, and D. Jha, "From SAM to DINOv2: Towards distilling foundation models to lightweight baselines for generalized polyp segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2026, *arXiv:2512.09307*.

[32] S. Agnihotri, S. Majhi, D. R. Nayak, D. Mahapatra, and D. Jha, "Induce to empower: Improving lightweight baselines via foundation model induction for generalized polyp segmentation," 2026, *arXiv:2607.17208*.

[33] S. Yang *et al.*, "SegDINO: Introducing multi-scale structure into DINO for efficient medical image segmentation," 2026, *arXiv:2606.17972*.

[34] K. Wang, J. H. Liew, Y. Zou, D. Zhou, and J. Feng, "PANet: Few-shot image semantic segmentation with prototype alignment," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, doi: 10.1109/ICCV.2019.00929.

[35] Q. Fan, W. Pei, Y.-W. Tai, and C.-K. Tang, "Self-support few-shot semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2022, doi: 10.1007/978-3-031-19800-7_41.

[36] E. J. Hu *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.

[37] Z. Zhong, Z. Tang, T. He, H. Fang, and C. Yuan, "Convolution meets LoRA: Parameter efficient finetuning for segment anything model," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[38] S. Jie and Z.-H. Deng, "Convolutional bypasses are better vision transformer adapters," 2022, *arXiv:2207.07039*.

[39] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 12179–12188.

[40] Q. Xu, R. Zhang, Y. Zhang, Y. Wang, and Q. Tian, "A Fourier-based framework for domain generalization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 14383–14392.

[41] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bull.*, vol. 1, no. 6, pp. 80–83, 1945.

[42] S. Holm, "A simple sequentially rejective multiple test procedure," *Scand. J. Statist.*, vol. 6, no. 2, pp. 65–70, 1979.

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